

UNIVERSITY NAME

*Rigid Body Mechanics — Final Exam**Year: 2015***Exercise 1:**

1. Show that the velocity vector field of a rigid body defines a torsor.
2. Show that the angular momentum is an antisymmetric vector field.
3. Show the following formula: for any point A belonging to the rigid body S of mass m and center of mass G ,

$$L_A(S/R_0) = \Pi_A(S) \cdot \hat{\Omega}(S/R_0) + m \cdot \overrightarrow{AG} \times \overrightarrow{V}(A/R_0).$$

4. (a) Write the theorem of angular momentum about a point A that is mobile in R_0 .
(b) Prove this theorem.

Answer Area

1. The velocity field of a rigid body S in a frame R_0 satisfies:

$$\vec{V}(P/R_0) = \vec{V}(A/R_0) + \vec{\Omega}(S/R_0) \times \overrightarrow{AP}$$

for any two points $A, P \in S$. This defines a *torsor* (a screw field) because:

- It is linear in the point P ,
- The difference between velocities at two points is given by the same $\vec{\Omega}$ for all P ,
- The field is equiprojective: $(\vec{V}(P) - \vec{V}(A)) \cdot \overrightarrow{AP} = 0$.

2. Angular momentum $\vec{L}_A(S/R_0)$ about point A is defined as:

$$\vec{L}_A = \int_S \overrightarrow{AP} \times \vec{V}(P) dm$$

Using the rigid body velocity field, one can show that for another point B :

$$\vec{L}_B = \vec{L}_A + \overrightarrow{AB} \times \vec{P}(S/R_0)$$

where $\vec{P}$ is the linear momentum. This transformation law is that of an antisymmetric vector field (or dual vector of a torsor).

3. Starting from definition:

$$\begin{aligned} \vec{L}_A(S/R_0) &= \int_S \overrightarrow{AP} \times \vec{V}(P/R_0) dm \\ &= \int_S \overrightarrow{AP} \times [\vec{V}(A/R_0) + \vec{\Omega} \times \overrightarrow{AP}] dm \\ &= \left(\int_S \overrightarrow{AP} dm \right) \times \vec{V}(A/R_0) + \int_S \overrightarrow{AP} \times (\vec{\Omega} \times \overrightarrow{AP}) dm \\ &= m \overrightarrow{AG} \times \vec{V}(A/R_0) + \mathbf{I}_A(S) \cdot \vec{\Omega} \end{aligned}$$

where $\mathbf{I}_A(S)$ is the inertia tensor at A .

4. (a) Theorem of angular momentum about a mobile point A :

$$\frac{d}{dt} \vec{L}_A(S/R_0) = \vec{M}_A(\text{ext}) + m \overrightarrow{AG} \times \vec{a}(A/R_0)$$

where $\vec{M}_A(\text{ext})$ is the moment of external forces about A .

- (b) Proof: Start from the fixed-point angular momentum theorem for A fixed in R_0 , then use the transport formula for the time derivative when A is moving.

Exercise 2:

A physical pendulum S , consisting of a homogeneous rod of length B_a and negligible mass, welded to a homogeneous ball of mass m , radius a , and center of mass G , is in motion relative to a fixed Galilean reference frame $R_0(X_0, Y_0, Z_0)$ with basis $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$.

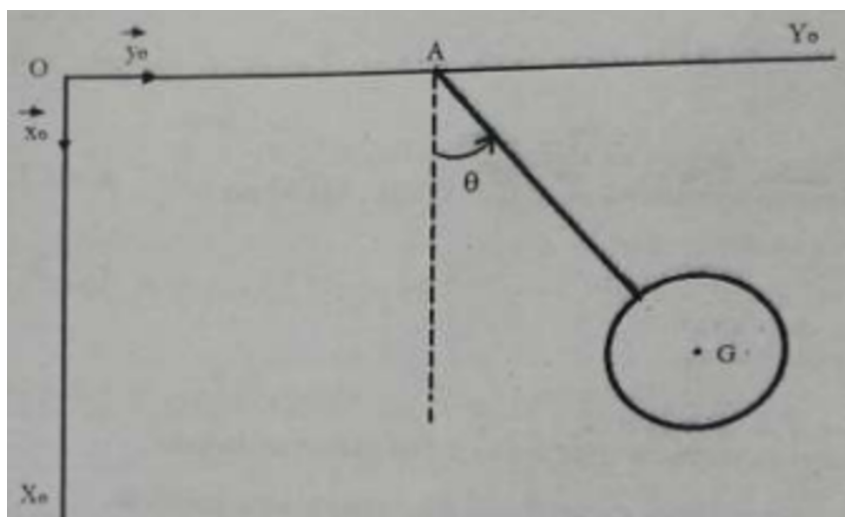
The system is moving in the vertical plane X_0OY_0 (with $G \in X_0OY_0$) and A is a mobile point. The motion is frictionless at A . R denotes the reaction force exerted on the system at point A .

The position of point A on the axis OY_0 is given by the coordinate y (with $\overrightarrow{OA} = y\bar{y}_0$) and the position of the system relative to the vertical is defined by the angle θ . The system is displaced from its equilibrium position and then released without initial velocity.

1. Calculate the inertia matrix of the ball at point G .
2. Determine the velocity vector of point G in R_0 .
3. By applying the dynamic resultant theorem, show that y and θ are related by the following relation: $y = -6a\dot{\theta} \cos \theta$. Deduce that the kinetic resultant of the system is given by:

$$P(S/R_0) = -6ma\dot{\theta} \sin \theta \bar{x}_0$$

4. By applying the kinetic energy theorem to the solid, determine the first integral of energy and deduce the differential equation of motion of the solid.
5. Retrieve the differential equation of motion of the solid by applying the angular momentum theorem to the solid, at point A , in the reference frame R_0 .
6. Determine the period of small oscillations (keeping only first-order terms in θ and $\dot{\theta}$).



Answer Area

1. For a homogeneous solid sphere of mass m , radius a , inertia matrix at G in principal axes:

$$\mathbf{I}_G = \frac{2}{5}ma^2 \mathbf{I}_3$$

where $\mathbf{I}_3$ is the 3×3 identity matrix.

2. Let O be the fixed pivot. The ball is welded to a massless rod of length Ba . Let G be the center of the ball. Then:

$$\overrightarrow{OG} = (Ba + a \sin \theta) \vec{x}_0 + a(1 - \cos \theta) \vec{y}_0$$

The velocity:

$$\vec{V}(G/R_0) = \dot{\theta} [(Ba + a \cos \theta) \vec{y}_0 + a \sin \theta \vec{x}_0]$$

(Derived from $\vec{V} = \frac{d}{dt} \overrightarrow{OG}$ with $\dot{\theta}$ as the rotation rate.)

3. Using dynamic resultant theorem: sum of forces = $m\vec{a}_G$. Forces: weight $mg\vec{y}_0$ and reaction $\vec{R}$ at A . Projecting along $\vec{x}_0$ and eliminating R_x yields:

$$y = -6a\dot{\theta} \cos \theta$$

The linear momentum:

$$\vec{P}(S/R_0) = m\vec{V}(G/R_0) = -6ma\dot{\theta} \sin \theta \vec{x}_0$$

(after using the constraint from no friction at A).

4. Kinetic energy:

$$T = \frac{1}{2}m\vec{V}_G^2 + \frac{1}{2}\mathbf{I}_G\dot{\theta}^2$$

Potential energy: $V = mg \cdot OG_y = mga(1 - \cos \theta)$. Energy conservation: $T + V = \text{const.}$ Differentiating gives:

$$\left(\frac{7}{5}ma^2 + m(Ba)^2 \right) \ddot{\theta} + mga \sin \theta = 0$$

5. Angular momentum theorem about A :

$$\frac{d}{dt} \vec{L}_A = \overrightarrow{AG} \times m\vec{g} + \overrightarrow{AA} \times \vec{R} = \overrightarrow{AG} \times m\vec{g}$$

Computing $\vec{L}_A = \mathbf{I}_A \dot{\theta} \vec{z}_0$ and differentiating yields the same differential equation as in (d).

6. For small θ , $\sin \theta \approx \theta$:

$$\ddot{\theta} + \frac{g}{L_{\text{eff}}} \theta = 0$$

with $L_{\text{eff}} = \frac{\frac{7}{5}a^2 + (Ba)^2}{a}$. Period:

$$T = 2\pi \sqrt{\frac{L_{\text{eff}}}{g}}$$